

Why Gauge Symmetry Cannot Be Ontic

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Abstract

Gauge symmetry is often described as a fundamental feature of physical reality. In this paper, we argue that this interpretation is untenable. Treating gauge symmetry as ontic—i.e. as a representation-independent structure that constrains physical possibilities—leads either to redundant constraint imposition or to unobservable surplus structure. We show that gauge symmetry must instead be understood as representational, functioning as a bookkeeping device within particular descriptive frameworks. The result is eliminative: no alternative ontology of gauge symmetry is proposed, only the conclusion that gauge symmetry realism is incoherent.

1 Introduction

Gauge symmetry occupies a peculiar position in modern physics. It is routinely described as “fundamental,” yet it is also widely acknowledged that different gauge choices correspond to the same physical situation. This duality—fundamental yet redundant—has long been noted but rarely pressed to a decisive conclusion.

This paper addresses a narrow but decisive question:

Can gauge symmetry be ontic?

By “ontic,” we mean a structure that exists independently of representation and that constrains physical possibilities rather than merely organizing descriptions. We argue that gauge symmetry cannot meet this criterion. Attempts to treat gauge symmetry as ontic either duplicate constraints already imposed by locality and consistency or introduce unobservable surplus structure that carries no physical content.

The argument is eliminative. We do not deny the utility of gauge symmetry, nor do we propose an alternative theoretical framework. We show only that gauge symmetry cannot coherently be interpreted as a feature of reality itself.

2 What Ontic Gauge Symmetry Would Require

An ontic structure must satisfy minimal criteria:

1. **Representation-independence:** it must not depend on arbitrary descriptive choices.
2. **Constraint-imposition:** it must restrict physical possibilities.

3. **Non-redundancy:** it must not merely restate constraints imposed elsewhere.

In this paper, “ontic” is used in a minimal structural sense: an ontic structure must impose genuine physical restrictions beyond those already enforced by other accepted constraints. The question at stake is whether gauge symmetry can play such a role.

If gauge symmetry is ontic, then gauge-related descriptions must reflect a real multiplicity or constraint in the world, not merely in representation.

3 Redundancy of Constraint Imposition

Gauge symmetry is often said to enforce consistency conditions in physical theories. However, any physically acceptable description must already satisfy locality and internal consistency. These constraints restrict how fields and interactions may be represented without invoking global coordination or unphysical degrees of freedom.

The problem is not redundancy per se, but redundant constraint imposition: a purported ontic structure that adds no new physical restriction cannot ground a realist interpretation.

If gauge symmetry is treated as ontic, two possibilities arise:

1. **Gauge symmetry enforces constraints already required by locality and consistency.** In this case, it is redundant. It adds no new structural restriction and therefore does not qualify as an invariant feature of reality.
2. **Gauge symmetry enforces additional constraints beyond locality and consistency.** In this case, it introduces surplus structure that has no observable consequences, since gauge-related descriptions are empirically indistinguishable.

Either way, gauge symmetry fails to play a non-redundant ontic role.

4 The Problem of Unobservable Structure

Ontic structures must have principled physical significance. Gauge symmetry, however, relates descriptions that are empirically indistinguishable. No observation can discriminate between gauge-related states.

This is not a contingent limitation but a structural one. If gauge symmetry were ontic, differences between gauge-related configurations would correspond to real differences in the world. The fact that no such differences can be observed implies either that gauge symmetry introduces physically meaningless distinctions, or that gauge symmetry does not correspond to ontic structure.

The first option is incompatible with any principled realism. The second option is eliminative.

5 Gauge Symmetry as Representational Bookkeeping

The difficulties above dissolve if gauge symmetry is treated as representational. On this view, gauge symmetry functions as a bookkeeping device that organizes local descriptions while avoiding illicit global constraints.

In particular:

- gauge redundancy allows local descriptions to be formulated flexibly,
- gauge transformations track equivalences between descriptions rather than physical differences,
- gauge symmetry ensures internal consistency without introducing new physical structure.

5.1 Does Structural Realism Rescue Gauge Symmetry?

One might attempt to rescue gauge symmetry by appealing to structural realism: even if gauge symmetries are not “things,” they might still be invariant relational structures.

This move does not succeed. Structural realism requires that the structure in question constrain physical possibilities. Gauge symmetry does not. It merely identifies equivalence classes of descriptions. Treating such equivalence relations as ontic structure reintroduces redundancy without explanatory gain.

Structural realism therefore does not salvage gauge symmetry realism; it merely restates the problem in relational language.

6 Consequences

The conclusion is not that gauge symmetry is dispensable, but that its role is representational rather than ontic. Gauge symmetry organizes descriptions; it does not describe reality.

The present argument is conditional on retaining locality and internal consistency as non-negotiable structural constraints. Under those conditions, gauge symmetry realism fails.

This result:

- does not depend on dynamics,
- does not rely on empirical considerations,
- does not presuppose a specific physical theory.

It arises solely from structural analysis of what an ontic symmetry would have to be.

7 Conclusion

Gauge symmetry cannot coherently be interpreted as an ontic feature of reality. Treating it as such leads either to redundant constraint imposition or to unobservable surplus structure. The only consistent interpretation is that gauge symmetry is representational: a tool for organizing local descriptions rather than a fundamental constituent of the world.

This conclusion is eliminative, not revisionary. It narrows the space of coherent interpretations without proposing a replacement ontology.